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Assessment of Street bikeability by Leveraging Multimodal Large Language Models with Geographical Context by Computer Vision

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Abstract

While conventional bikeability research relies on objective street attributes, a growing body of literature highlights a significant discrepancy between objective metrics and human perception, and we need to consider perceived bikeability besides urban environment. Bikeability assessment requires capturing not only objective street-level features but also human perception to have better bikeability assessments. Computer vision algorithms excel at precise pixel-level quantification of environmental attributes, but they cannot provide subjective perceptions. Multimodal large language models (MLLMs) offer holistic scene understanding that may simulate human perception, but lack the spatial quantification of street environment. This study bridges this gap by proposing an integrated framework that leverages the complementary strengths of both approaches. A structured Geographical Context Database—synthesizing semantic segmentation metrics and object detection outputs using computer vision techniques, and points-of-interest data—is constructed to ground MLLM reasoning in measurable environmental attributes, enabling more context-sensitive bikeability evaluation. Validation against human assessments confirms that this integration substantially improves its alignment with human perceptions compared to image-only evaluation by MLLM. Using an adaptive pairwise comparison system, we translate MLLM judgments into quantitative scores across three dimensions: Accessibility, Safety, and Comfort. Beyond producing bikeability scores, this study further exploits the reasoning texts generated during pairwise comparisons as an analytical resource. Analyzing the reasoning texts, we find that the factors the model mentions are often those that most influence its judgments, indicating that its stated reasons are consistent with its decisions. These findings position a context-grounded MLLM as a scalable proxy for perceived bikeability. These findings offer actionable insights for cycling-oriented urban design by identifying high-leverage intervention targets overlooked by traditional assessments.

Keywords

Street bikeability; Multimodal Large Language Models (MLLMs); Computer vision; Street view images

1 Introduction

Cycling as active travel is widely recognized as contributing to people's health and well-being (Millard-Ball et al., 2025). Bikeability, broadly defined as the extent to which an environment supports safe and convenient cycling, plays a pivotal role in influencing cycling behavior (Ito et al., 2021; Li et al., 2017). This concept has gained prominence in urban planning and transportation research as a tool to evaluate how built environments shape travel behavior, mode choice, and active mobility promotion (Ito et al., 2021; Xu et al., 2025). Traditionally, bikeability has primarily been assessed through the street-level features that directly shape cyclists' route choices and experiences. Recent advances in street view imagery and computer vision (CV) algorithms—particularly semantic segmentation and object detection—have enabled the automated quantification of many objective features, such as the proportion of street greenery, building enclosure, and bike lane types, facilitating numerous studies investigating their influence on cycling (Xu et al., 2025).

Despite these advancements, current CV-based approaches face two critical limitations that hinder comprehensive bikeability assessment. First, off-the-shelf CV models trained on benchmark datasets struggle to detect the full range of cycling-relevant features. Existing studies often assess bikeability using only a limited subset of features, overlooking other attributes that cyclists consider important such as bike lane segregation and road types. Second, and more fundamentally, CV models evaluate features in isolation through pixel-level quantification (e.g., proportion of greenery), but cannot determine whether these elements collectively create a holistically safe or pleasant cycling environment. The spatial arrangement and contextual harmony among features—which critically shape subjective perceptions—remain beyond conventional CV capabilities. As a result, comprehensive perceptual assessments still rely heavily on costly manual audits or crowdsourced surveys (Huang et al., 2025).

Recently, multimodal large language models (MLLMs), particularly those with advanced visual functions such as Gemini and Qwen-VL, have introduced a transformative approach to image analysis by simulating human perception and reasoning (Ki et al., 2025; Zhang et al., 2025). It is possible to use MLLM model and analyze human perception with Human-AI Symbiosis

framework. Unlike traditional CV models that rely on pre-defined categories and pixel-level classification, MLLMs can interpret visual scenes in a more holistic and context-aware manner, identifying subtle environmental cues and assessing their combined influence on user experience.

Unlike unimodal models that process single data types, MLLMs represent a significant advancement by integrating multiple data modalities such as texts, images, and geospatial information into traditional LLMs (Luo et al., 2025; Lu et al., 2025). Previous studies have shown the potential that inputting GIS data into MLLMs can help enrich the evaluation context to complement the limited access of street view imagery to broader spatial context (Ki et al., 2025). In this study, we integrate MLLM-based image analysis with structured geographical context by CV to support more accurate and consistent bikeability assessment.

In this study, street bikeability refers to perceived suitability of a specific street segment for cycling, as assessed from the cyclist’s street-level visual perspective and its associated geographical context around the cyclist. China has a long-established bicycle culture, but has been losing bicycle travel shares to other transport modes in cities. Using central Shanghai as a case study, this research addresses three key questions: (1) To what extent does geographical context extracted by CV enhance the ability of MLLMs to capture subjective bikeability assessments and align with human perceptions? (2) Do the street factors an MLLM most often mentions correspond to those that most influence its bikeability judgments? (3) How do the same street environmental factors exert dimension-dependent effects across accessibility, safety, and comfort? We posit that through combining MLLM’s human-like reasoning with geographical context by CV, our framework facilitates more nuanced and human-centric bikeability evaluations.

2 Literature Review

2.1 Objective street environment and perceived bikeability measurement

Most current research quantifies bikeability through objective street environmental attributes, such as bike facilities, intersection density, slope of bike routes, and land-use mix, often measured through tools like the Bikeability Index or GIS-based assessments (Ma et al., 2017; Ito et al., 2021). Street-level built environment elements—such as greenery, building enclosure, and bike lane type—not only represent objective spatial configurations but also significantly affect residents’ subjective perceptions. Common findings indicate that presence of dedicated bike lanes, higher

road network density and land-use diversity significantly enhance cycling experiences (Kosmidis et al., 2024; Ma et al., 2024; Zhao et al., 2017). Street greenery is one of the most frequently studied visual indicators. Research shows that street view images can reflect individuals' perceived greenness (Wang et al., 2020). A significant positive correlation was found between objectively measured street greenery and perceived cycling safety (Ye et al., 2024). Building enclosure, defined as the degree to which buildings visually surround a street, has been identified as an important spatial variable associated with cyclists' sense of spatial control and psychological safety. Streets with higher enclosure are generally perceived as safer and more supportive of cycling (Xu L et al., 2026). Bike lane type is another critical determinant. Physically separated bike lanes—such as those delineated by fences, curbs, or vegetation—are positively associated with cycling uptake, while mixed-traffic lanes shared with motor vehicles are often perceived as unsafe (Ji et al., 2022).

Recently, a growing body of literature highlights the discrepancy between objective and perceived bikeability, suggesting the need to incorporate human perceptual factors into bikeability assessments. Perceived bikeability encompasses subjective evaluations of these environment attributes (Qiu et al., 2023; Ogawa et al., 2024; Qi et al., 2025). Perceived bikeability can be adapted into a tiered structure: Accessibility, Safety, and Comfort (Reggiani et al., 2022; Xu et al., 2025). Accessibility reflects the ease of reaching needed and desired destinations around (Yang et al., 2019). Safety concerns personal security and the risk of traffic conflicts, strongly shaped by road design and operations (Ferenchak et al., 2025; Kaplan et al., 2018). The core element is the type of separation between motor vehicles and bicycles—for example, physical barriers, painted separation lines, or no separation. Comfort synthesizes physical and psychological sensations during the ride, reflecting environmental friendliness and sensory ease. Key affecting factors include appropriate greenness and enclosure (Zhang et al., 2024).

Perceived bikeability often mediates the relationship between objective conditions and actual behavior. Xu et al. (2026) revealed that both objective street features and subjective perceptions are important. Studies have shown that well-maintained bike lanes and visible greenery can improve perceived safety and aesthetic appeal of cycling (Harvey et al., 2015). For instance, Gan et al. (2021) revealed that perceived bikeability was significantly associated with cycling, highlighting the importance of subjective evaluations beyond macro-scale objective metrics alone. Higher green coverage contributes not only to an enhanced sense of comfort, but also to increased

frequency of cycling trips (Zhang et al., 2024). However, research that examines the relationship between objective built environment and perceived bikeability measurements is still limited.

Street View Images (SVIs) provide photographs of streetscapes closely matching a cyclist’s visual perspective of the road environment (Rzotkiewicz et al., 2018), and show distinct advantages in extracting fine-grained objective environment features, including buildings, physical separation from vehicles, and greenery characteristics (Larkin et al., 2021). Scholars have also utilized SVIs with human ratings or crowdsourced surveys to collect perceptual preference data (Wang et al., 2019; Fitch et al., 2022; Koo et al., 2022). For instance, pioneering datasets such as Place Pulse introduced SVIs-based online pairwise voting methods to measure perceptions of safety, class, and uniqueness, where participants respond to questions like “Which place looks safer/more comfortable?” (Saleses et al., 2013). However, reliance on existing datasets raises concerns regarding cross-regional applicability. The dataset does not include SVIs from mainland China; consequently, streetscape perception analyses based on this open-source dataset may face questions about the generalizability and contextual applicability of the resulting models. Few studies have constructed subjective perception training datasets. For instance, Huang and Oki (2025) have conducted pairwise comparisons of SVIs to train deep convolutional neural networks (DCNN) that predict perceptual scores for each image, with predictions validated against survey results to assess accuracy. However, constructing subjective perception training datasets remains labor-intensive in deep learning based SVI studies (Zhang et al., 2018).

2.2 Potential of multimodal large language models in bikeability assessment

Large language models (LLMs) have been widely adopted for natural language data processing in planning and urban analytics, including tasks such as planning policy identification and text mining for urban problem sensing. In particular, Wang et al. (2026) demonstrate that LLMs can achieve accurate and stable performance in sentence-level text classification settings, suggesting that LLMs can be used as reliable classifiers under limited annotation budgets. Prior studies have applied language-based methods to derive semantic themes and sentiment signals, highlighting the capability of LLMs to operationalize abstract concepts from texts (Osorio-Arjona et al., 2021; Gugulica et al., 2025).

Advancements in multimodal large language models (MLLMs), are opening new avenues for assessing bikeability and other subjective urban qualities at scale. MLLMs represent a significant

advancement by extending traditional LLMs by integrating multiple data modalities such as text, images, and geospatial information into a unified framework (Luo et al., 2025). Unlike unimodal models that process single data type (e.g., text-only LLMs or image-only DCNN model), MLLMs leverage cross-modal learning to generate richer, more contextualized outputs, enabling them to simulate human-like reasoning across diverse inputs (Lu et al., 2025; Crooks et al., 2024). Furthermore, their transformer-based architecture provides context-aware mechanisms for global understanding and relational reasoning, overcoming limitations in traditional methods that neglect element interactions. These characteristics align closely with the perceptual nature of cycling experience, because cyclists evaluate streets by jointly interpreting visible environmental cues, such as bike lanes, greenery, traffic conflicts, and enclosure, together with non-visual contextual information, such as nearby destinations.

One of the most promising applications of MLLMs lies in their transformative image vision reasoning capabilities, which mimic human perception (Crooks et al., 2024). Models with its multimodal processing of text and images can perform advanced scene understanding. In urban planning and geography, integrating MLLMs with SVIs has shown potential for gaining deeper insights into built environments (Fu et al., 2025). Recent studies have applied MLLMs with visual-reasoning capabilities to assess urban safety based on SVIs, demonstrating strong alignment with human subjective evaluations (Zhang et al., 2025). Similarly, MLLMs have demonstrated the ability to mirror human aesthetic evaluations of natural landscape scenes across multiple preference dimensions (Tung et al., 2025; Wu et al., 2026). This presents a promising direction for efficient and large-scale urban environment assessments without extensive human annotation.

Some studies have utilized MLLMs to evaluate streetscape characteristics such as visual quality and attractiveness, substantiating their effectiveness in capturing complex and subtle attributes in an automated manner (Malekzadeh et al., 2025). Previous work has examined whether MLLMs can approximate human perceptions of street walkability, suggesting that these models can serve as scalable evaluative tools for large-scale urban perception assessment when their outputs are carefully validated against human judgments (Ki et al., 2025; Wedyan et al., 2025).

Despite these advancements, challenges persist in applying MLLMs to evaluating urban environment including bikeability. First, while current MLLMs excel in general perception tasks, they often lack integration of domain-specific geographical context. Without incorporating structured geospatial data (e.g. detailed road attributes, nearby amenities, or greenery), the models

may miss critical cues that influence context-specific cycling experience. Second, much of the current research has focused on walkability metrics, whereas bikeability entails distinct needs in terms of infrastructure and use patterns such as dedicated cycling facilities, separation from traffic, road design, and trip duration, which differ markedly from pedestrian requirements (Xu et al., 2025).

To address these limitations, this study aims to advance bikeability evaluation by integrating MLLM-based image analysis with geographical context data, enabling more comprehensive and context-sensitive assessments of bikeability that reflect subjective perceptual cues.

3 Study area and data processing

3.1 Study area

The inner urban area of Shanghai (within Inner Ring Road), which spans 156 km², is selected as the study area (**Figure 1**), due to the large bike-sharing users in this area. Shanghai hosts the world’s largest bike-sharing system. The compact urban form makes the inner urban area a suitable case to evaluate the effects of street environment on perceived bikeability.

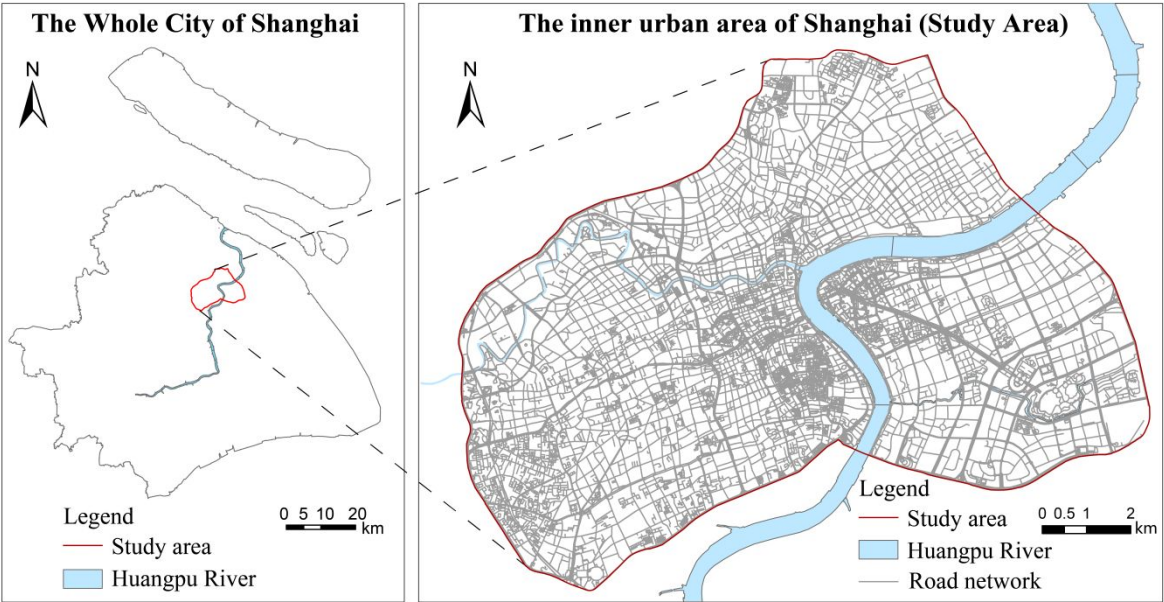


Figure 1. The inner urban area of Shanghai

3.2 Street view images data collection

SVIs were collected through the Baidu Maps API, and sampling points were generated at intervals of 50 meters. We finally acquired 7,108 images after data cleaning. 400 representative SVIs that best represent the spatial characteristics of Shanghai streets were randomly sampled to form our anchor set for subsequent manual scoring evaluation.

3.3 POI data processing

The POI data were sourced from Amap (Gaode Maps), providing a geographic context to supplement the MLLM in evaluating street-level cycling accessibility. We selected POI categories closely associated with daily cycling activities including commercial, office, shopping, culture, sports, education, and transportation processing. POIs were spatially matched and aggregated based on the coordinates of the corresponding SVIs. A 500 m buffer zone—representing a typical "last-mile" cycling catchment area—was established around each location. Within these buffers, the quantity and density of specific destinations were calculated.

4 Research methodology

This study develops an integrated analytical framework to assess street-level bikeability through street view imagery and geographical contextual data around the cyclists. The framework comprises four main components: (1) Derivation of bikeability indicators, (2) MLLM-based perceived bikeability assessment with geographical context derived from CV techniques, (3) Pairwise Comparison into continuous bikeability ratings, and (4) Validation of bikeability ratings against human evaluation, and analysis of MLLM reasoning texts (**Figure 2**).

First, we conceptualize bikeability as the suitability of a specific street segment for cycling, evaluated directly through the lens of a cyclist's perception across three distinct dimensions: Accessibility, Safety, and Comfort.

The methodological core is an iterative human–AI assessment loop that integrates qualitative perception modeling with quantitative ranking. The SVIs and associated geographic context derived from the CV model are the input layer. Through targeted prompt engineering, a MLLM acts as the elicitation mechanism to generate discrete pairwise preferences. These preference judgments are continuously converted into continuous scores using a pairwise rating system. This closed-loop mechanism dynamically recalculates and updates SVI scores until they reach stability, significantly reducing the sampling overhead required by exhaustive pairwise comparisons.

Finally, the MLLM assessment results are validated against human perceptions to ensure alignment. Furthermore, we perform a text analysis on the MLLM's generated reasoning paths to extract structured environmental factors, allowing us to model exactly how specific street-level attributes influence perceived bikeability.

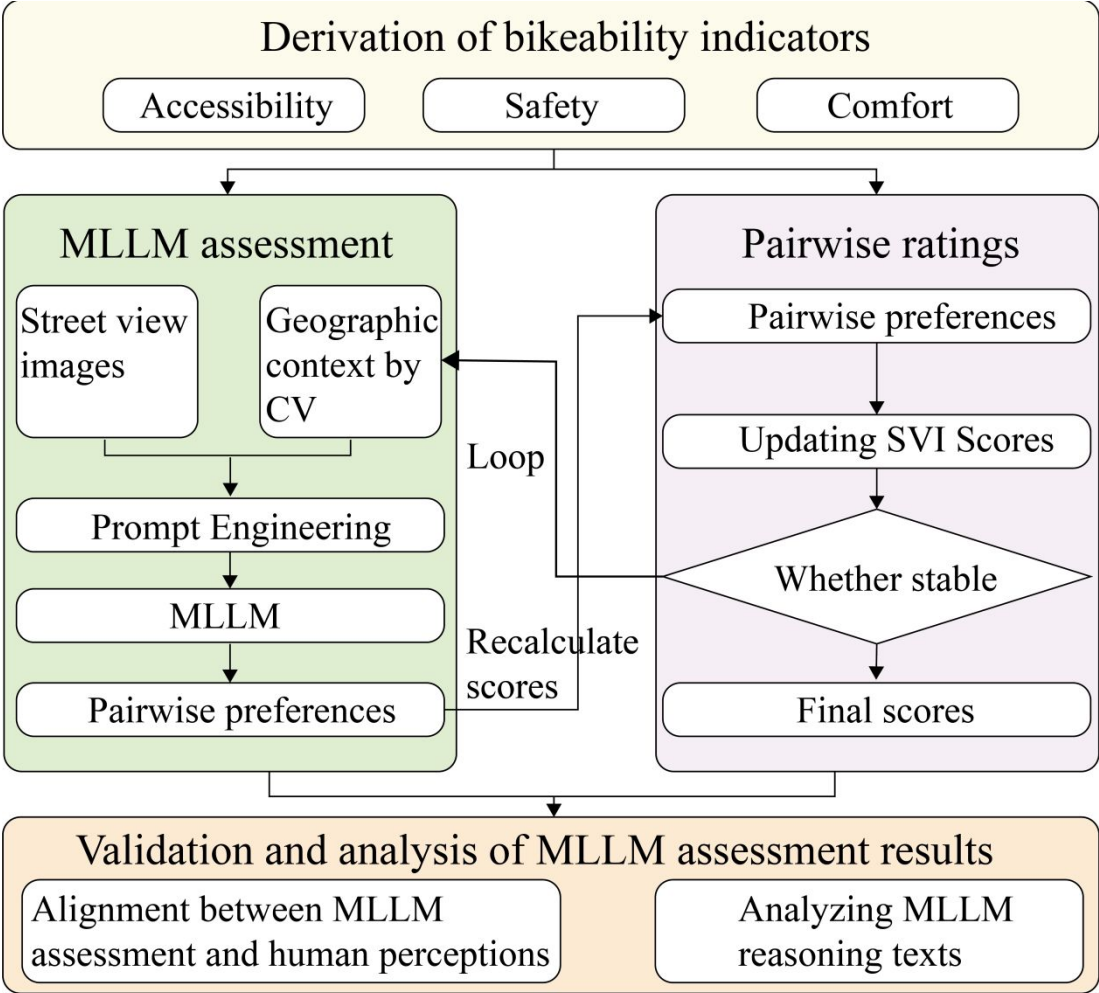


Figure 2. Analytical framework

4.1 SVIs semantic segmentation and object detection

This study examined a quantitative framework for assessing street environment variables that influenced street bikeability. The semantic segmentation of SVIs using the DeepLabV3+ neural network model was employed to extract the spatial features using SVIs (Zhong et al., 2025), which was pre-trained based on the Cityscapes dataset, allowing us to extract the pixel-level proportions of visual elements (e.g., buildings, vegetation). Based on the semantic segmentation results, we calculated objective visual indicators to support perceived bikeability assessment. The Green View

Index (GVI) was calculated as the proportion of vegetation pixels within each street view image, while the Sky View Factor (SVF) and enclosure ratio were derived from the proportions of sky, building, and other relevant morphological classes. These indicators were used primarily to inform Comfort assessments.

Cycle lane types, including protected by fences and green belts, striped, and mixed-traffic lanes, were identified through object detection and used to support Safety judgments. The dataset for model training was labeled by participants who had cycling experience. The object detection by YOLOv8 (Dai et al., 2025) was utilized to extract features from the street view images, to better capture critical information about vehicle-bicycle separation types present in the SVIs. Each cycle lane type includes approximately 400 images, divided in a 7:2:1 ratio for training, validation, and testing.

4.2 MLLM perception integrated with Geographical Context Database

Existing studies often fed SVIs directly into MLLMs to evaluate urban streetscapes. However, the output quality of large models was affected by the provided urban spatial context. We constructed a Geographical Context Database by integrating semantic segmentation and object detection features extracted from SVIs together with Points of Interest (POI). These features served as objective indicators of street environment for cyclists. (1) Morphological metrics derived from semantic segmentation, including GVI, SVF, and enclosure ratio, to support Comfort evaluations; (2) Infrastructure attributes from object detection such as bike lane separation types to support Safety evaluations; and (3) POI density to support Accessibility evaluations. The Geographical Context Database offered richer and more comprehensive contextual information for MLLM-based evaluation and thereby enhanced the accuracy of street bikeability assessments. The contrast between our analytical framework and existing works is shown in **Figure 3**.

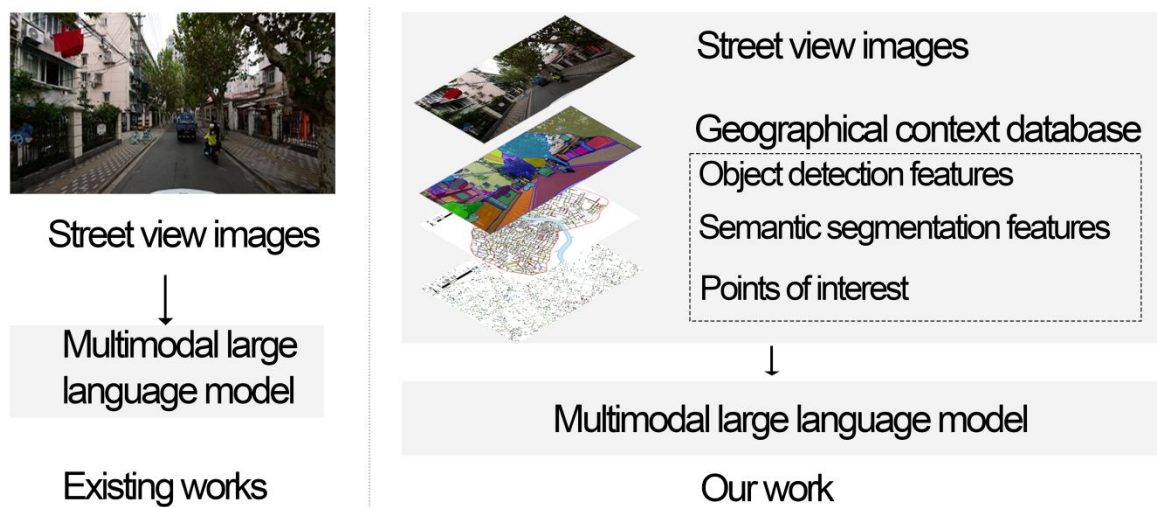


Figure 3. Existing works vs. our analytical framework.

The rationale for this multimodal integration stems from the complementary strengths of computer vision and MLLMs in processing geographical context information. Semantic segmentation models excel at extracting precise, pixel-level quantitative measurements from street view imagery (Ito and Biljecki, 2021). However, these models cannot assess whether quantified features collectively create a safe or comfortable cycling environment, as this requires holistic scene understanding and contextual reasoning (Ki et al., 2025). Conversely, while MLLMs demonstrate strong capabilities in interpreting visual scenes holistically, they encode images as patch-level visual tokens rather than preserving pixel-level accurate spatial information. For instance, when comparing images with subtle quantitative differences—such as streets with 35% versus 42% green coverage—MLLMs may produce vague semantic assessments (e.g., "both streets have trees") rather than recognizing meaningful environmental distinctions. CV and MLLM are therefore complementary — CV-derived semantic assessments provide what the MLLM cannot measure precisely, and the MLLM supplement urban perceptions.

By integrating geographical context as structured quantitative inputs, we enable the MLLM to ground its holistic reasoning in interpretable, measurable environmental attributes. We employ a structured text input approach, dynamically embedding fields from attribute tables into prompts as text, which can better help MLLM fetch required geographical context from dataset and reduce image perception ambiguity. This approach bridges the link between precise geospatial measurements and nuanced perceptual judgments.

4.3 Pairwise comparison to acquire perceived bikeability

Pairwise comparison is widely recognized for its robustness in visual perception research, offering higher inter-rater reliability than conventional Likert ratings (Gu et al., 2025). The Elo method is particularly suited for static visual datasets (Wu et al., 2025). First, its deterministic update rule aligns naturally with the fixed physical characteristics of street view images, which remain stable during the evaluation process. Second, Elo's independent scoring structures can be maintained across multiple perceptual dimensions without cross-dimensional interference.

We employ an Elo rating system to convert MLLM pairwise preferences into quantifiable bikeability scores across three dimensions: Accessibility, Safety, and Comfort. All images are initialized at a rating of 1000. Elo ratings E_A and E_B are calculated using the following formulas:

$$E_A = 1/(1 + 10^{(R_B - R_A)/400}) \quad E_B = 1/(1 + 10^{(R_A - R_B)/400}) \quad (1)$$

Where R_A and R_B represent the current ratings, E_A and E_B are the expected outcomes; 400 is an empirical scale factor controlling sensitivity to rating differences (Wu et al., 2025). After each comparison, ratings are updated based on the actual outcome and the expectation deviation:

$$R'_A = R_A + K(S_A - E_A) \quad R'_B = R_B + K(S_B - E_B) \quad (2)$$

After a comparison (a “match”), the ratings are updated using Elo's update formula. S_A is the actual outcome for item A, and K is a constant step size controlling the update magnitude. The term $(S_A - E_A)$ represents the surprise of the outcome, for example if A wins when it was expected to lose, its rating increases significantly, whereas an expected win yields only a minor increase.

We implement an adaptive closed-loop system that dynamically prioritizes informative comparisons rather than exhaustive pairwise comparisons (**Figure 4**). The first step is selective image pairing. At the start of each evaluation round, the system retrieves all images within the current bikeability dimension and examines their current Elo ratings. Instead of random pairing, the algorithm identifies images with similar ratings. This strategy is to focus computational resources on comparisons that genuinely refine the ranking structure, avoiding redundant evaluations between images with significant ranking disparity.

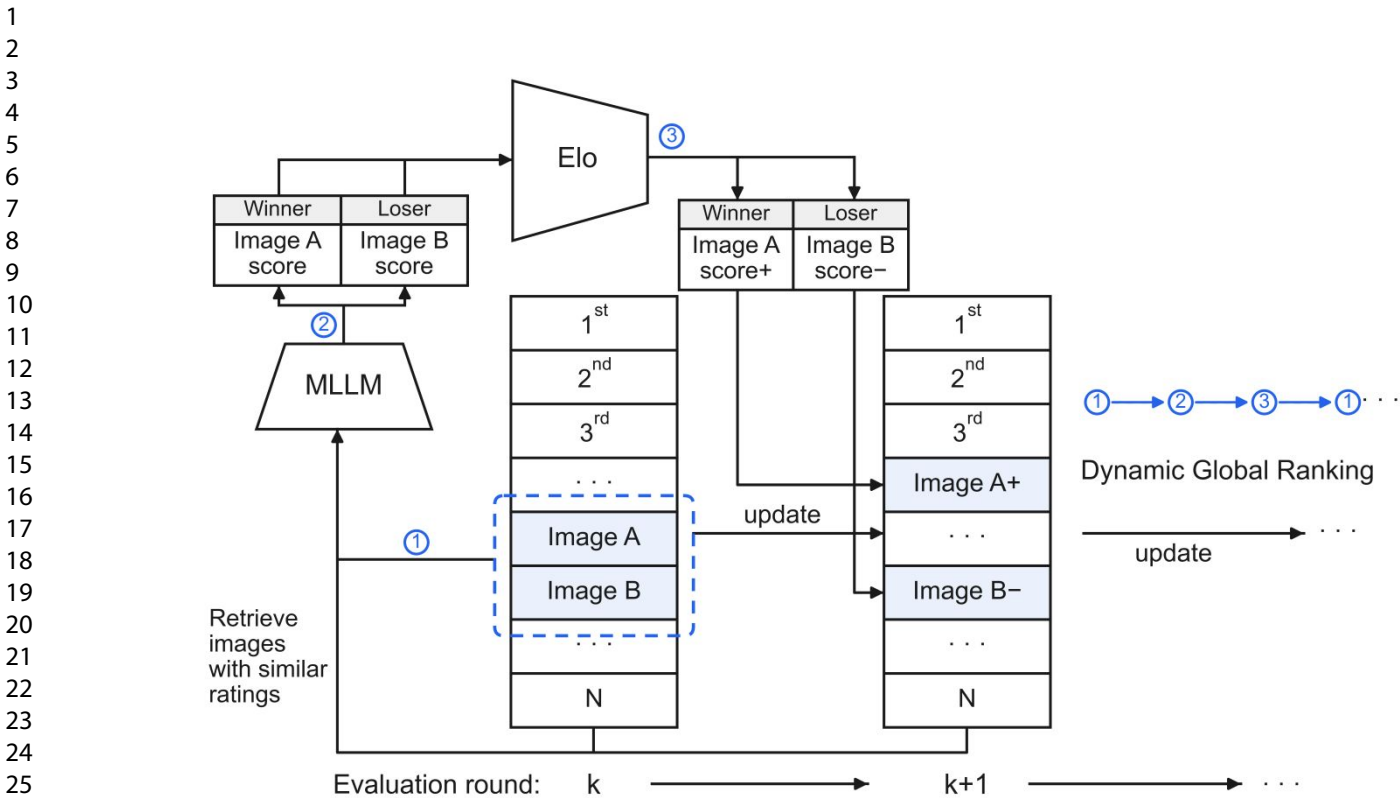


Figure 4. A closed adaptive loop for the MLLM pipeline evaluation.

The second step is for the MLLM to perform win-loss evaluation. Selected image pairs are combined with their corresponding geographical context. The MLLM is constrained through prompt engineering to produce structured outputs containing both a binary win-loss decision and supporting textual reasoning.

The third step is to calculate Elo scores and update rankings based on win-loss results. As illustrated in Figure 4, the comparison outcome from MLLM is then fed into the Elo mechanism: the winner (Image A) received a rating increase (score+) while the loser (Image B) received a rating decrease (score-). Critically, this update propagates through the entire ranking structure. As shown in the transition from round k to round $k+1$, Image A ascends in the ranking hierarchy while Image B descends, consequently reshuffling the relative positions of all images between and below them.

This architecture creates a self-reinforcing feedback loop: early rounds resolve large rating disparities quickly, while later rounds automatically concentrate on images clustered at similar rating levels—precisely where ranking uncertainty remains highest. The iterative process continues until convergence is achieved, defined as the point where the average absolute rating change per image falls below a predetermined tolerance threshold across consecutive rounds. This

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3 criterion ensures that the ranking has stabilized sufficiently for downstream analysis while
4 avoiding computational expenditure on marginal refinements. In practice, this adaptive approach
5 may substantially reduce the number of comparisons needed comparing to complete pairwise
6 enumeration while maintaining ranking quality, making it scalable to large urban street networks.
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10 **4.4 Quantifying the salience and influence of street factors on bikeability perception**

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12 The reasoning texts generated during pairwise comparisons constitute a rich but unstructured
13 corpus detailing why the MLLM preferred one streetscape over another. Rather than providing
14 exhaustive descriptions, these texts selectively depict specific elements of each image to justify
15 the model's win-loss judgments. Once structured into explicit, factor-level annotations, this
16 qualitative reasoning becomes amenable to statistical modeling. This approach allows us to
17 separate how often a factor is noticed from how strongly it impacts the decision. To achieve this,
18 we developed a two-stage analytical pipeline: structured factor extraction followed by joint
19 statistical modeling of factor salience and influence. This framework transforms qualitative
20 MLLM reasoning into interpretable, statistically grounded measures that ultimately uncover the
21 underlying principles governing bikeability perception.
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30 To capture built environment factors mentioned in the text for subsequent statistical and
31 quantitative analysis, we employed Qwen-VL-7B-Instruct to extract structured factor information
32 from pairwise comparison records. We first developed a standardized factor dictionary, each factor
33 entry specifies a canonical code, data type, and dimension-specific polarity indicating whether
34 higher values represent positive or negative influences on bikeability.
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39 The extraction procedure used carefully engineered prompts that instructed the model to: (a)
40 identify all environmental factors mentioned in the reasoning texts, (b) determine which image
41 side (A or B) each factor description pertains to, (c) identify which side holds the advantage for
42 each factor.
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46 With structured factor records in hand, we quantified street factors along two complementary
47 dimensions including salience and influence. Factor salience captures how frequently each factor
48 appears in the MLLM's reasoning, measured as the proportion of comparison records in which a
49 given factor is mentioned, while factor influence captures how strongly each factor shapes the final
50 preference outcome.
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54 To estimate the influence of each factor, we employed a paired-comparison logistic regression
55 framework that jointly models all factors while accounting for their correlations. For each pairwise
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comparison between Image A and B, we define $Y = 1$ if Image B wins and $Y = 0$ otherwise. The log-odds of Image B winning, $(\text{logit}(P(Y = 1)))$, is modeled as a linear combination of factor advantage indicators:

$$\text{logit}(P(Y = 1)) = \beta^0 + \sum(\beta_f \cdot \Delta f) \tag{3}$$

where Δf is a binary indicator equal to 1 if factor f favors Image B, and 0 if it favors Image A. Each coefficient β_f represents the change in the log-odds of winning associated with an advantage in factor f , holding all other factors constant.

Given the potential multicollinearity among factor indicators, we estimate Equation (3) using an L2 (ridge) penalty to yield stable coefficients. However, a given β_f does not correspond to a fixed change in win probability. Instead, the same coefficient shifts the predicted probability substantially in closely matched comparisons, but only slightly in one-sided ones.

To address this, we summarize each factor's influence using its average marginal effect (AME). For every comparison, the fitted model predicts the probability of Image B winning by first assigning the advantage for factor f to Image B, and then to Image A. The AME is the mean difference between these paired predictions across all comparisons. This approach translates β_f onto an intuitive probability scale, making it directly comparable across factors. Finally, we report the AME alongside the frequency with which each factor distinguishes the two street views.

5 Results and Findings

5.1 Distribution of cycling lane types

The object detection model achieved classification accuracy of 0.78, 0.91, 0.80, 0.84 for lanes protected by fences, green belts, striped cycle lanes, and mixed-use roads, demonstrating its effectiveness in detecting cycling lane separation types. Figure S1 shows A substantial share of streets lacks physically separated cycling infrastructure, which may weaken people's perceived traffic safety and, in turn, reduce their intention to cycle. To simplify the analysis of cycling infrastructure, the categories "protected by fence" and "green belt" are combined into a single category of physically segregated lanes. This consolidation reduces the total number of classifications from four to three.

5.2 Alignment between human perceptions and MLLMs in bikeability assessment

It is necessary to validate the alignment between MLLM and human perceptions in bikeability assessment. We have decomposed the ambiguous concept of "bikeability" into multiple more concrete dimensions including accessibility, safety, and comfort. The human perception pathway involved crowdsourced evaluation to establish preference scores that capture subjective bikeability assessments from these three dimensions. A total of 400 SVIs were manually selected based on location distribution and image quality to ensure spatial representativeness and visual clarity. An ELO rating algorithm was implemented to establish a scoring system, where 46 participants with cycling experience conducted approximately 24,000 pairwise comparisons across the three dimensions to serve as ground truth for evaluating MLLM alignment.

Volunteers were randomly assigned into two groups to independently evaluate and score the selected representative SVIs. The pairwise comparison results obtained from the volunteers were converted using the Elo rating algorithm. Based on the resulting Elo scores, all images were ranked, with higher scores indicating that a street scene is generally perceived by volunteers as offering better cycling quality under the corresponding evaluation dimension.

To further assess the consistency and reliability of the human evaluations, a comparative analysis was conducted between two rounds of independently performed Elo-based scoring (Table 1). The results reveal significant positive correlations across all three dimensions—accessibility, safety, and comfort ($p < 0.001$). This indicates a high level of agreement and stability in the human assessments, thereby providing a reliable foundation for subsequent model validation. Using these representative SVIs, we further compared human perception scores with assessments generated by the MLLM.

Table 1 Assessment of inter-rater reliability

Dimension	Pearson's r	Intraclass correlation coefficients	Pairwise Accuracy
Accessibility	0.621***	0.557	0.722
Safety	0.513***	0.464	0.668
Comfort	0.466***	0.427	0.674

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

The Pearson correlation coefficients (r) range from 0.466 to 0.621, suggesting moderate to strong linear associations between the two rounds of evaluation. The intraclass correlation coefficients further indicate moderate reliability, reflecting a reasonable degree of consistency in absolute ratings across evaluators. In addition, the pairwise accuracy values (0.668–0.722) demonstrate that the relative ranking of SVIs is preserved to a substantial extent between the two rounds. Notably, Accessibility again achieves the highest pairwise accuracy, implying that this dimension may be more intuitively and consistently perceived by participants. Overall, the convergence of multiple reliability metrics provides strong evidence that the human evaluation results are internally consistent, thereby supporting their use as a credible benchmark for subsequent comparisons with MLLM-generated assessments.

These human preference samples combined with corresponding geographical contextual were then used in the MLLM, enabling the model to develop more human-like scoring capabilities. Table 2 compares results between the MLLM’s and human’s evaluations. We employed the Pearson correlation coefficient to assess the consistency between human and MLLM perceptions. Results showed strong positive correlations and highly significance. It’s evidence to prove that the MLLM have the capability to reflect human perception to some extent.

We strengthened MLLM-based image analysis with structured geographical context information— including semantic segmentation outputs (e.g., green view index, enclosure ratio), object detection results (e.g., bike lane separation types), and POI data—to support more accurate and consistent bikeability assessments. By explicitly supplying geographical context for each image, the model’s judgement then might hinge on numeric comparison—a mode for which current LLMs were typically well trained, then we found the accuracy improved markedly. Context grounding raises the correlation between our model and human scores for three dimensions, relative to image-only evaluation. The results confirmed that this integration yields substantially higher alignment with human assessments compared to image-only evaluation.

Table 2 Comparison between MLLM-based bikeability and human assessments

Bikeability	Without Geographical context	With Geographical context	Δr
Accessibility	0.542***	0.664***	+0.122
Safety	0.385***	0.427***	+0.042
Comfort	0.301***	0.449***	+0.148

Figure S2 presents an MLLM-generated bikeability evaluation example of street view images. The MLLM was constrained through prompt engineering to produce structured outputs containing both a binary comparative judgment and supporting textual reasoning. This dual-output design—quantitative decision paired with qualitative explanation ensured traceability and interpretability: each ranking outcome could be traced back to specific environmental attributes and their quantitative values. These reasoning texts reflect the MLLM's attention to the street factors from a cyclist's perspective across specific dimensions.

5.3 Spatial distribution of Bikeability scores in different dimensions

After obtaining MLLM-generated bikeability scores for each street view image across the three dimensions, we revealed the spatial distribution of average bikeability scores (**Figure S3**). Accessibility scores are highest in core commercial and business districts, where a dense mix of destinations enhances reachability by bicycle. Safety scores are generally higher along arterial and secondary roads that feature clear physical separation between motorized and non-motorized traffic. Comfort scores are highest in districts characterized small blocks, dense street networks, and well-enclosed street spaces.

5.4 Street factor mention rate and its influence on bikability perception

Recent studies using MLLM evaluations often focus on the produced numerical scores and neglect the explanatory textual rationales of evaluation results. Our study exploited these explanatory outputs to understand the logic in MLLM judgments. We compare each factor's Mention Rate with its influence on bikability perception, measured as the average marginal effect (AME) on the predicted outcome, to test whether the cues the model articulates match those that drive its verdicts. AME reflects the mean change in the predicted probability that an image wins when a factor's advantage is switched from one side to the other. By comparing factor Mention Rate (from frequency analysis) with factor Influence Coefficient (from paired-comparison logistic regression model coefficients), we can identify perception gaps: factors that were underestimated (rarely mentioned but highly influential when present) and factors that were overestimated (frequently discussed but weakly decisive).

Table 3 reports each factor's Mention Rate, its influence expressed as the AME with a 95% Confidence Interval (CI), and the Odds Ratio of leading factors that have significant impact on

bikability perception across three dimensions. The corresponding Odds Ratio which is exp (Influence Coefficient (β_f)) restates the same coefficient on a multiplicative scale, as how many times greater the odds of winning are when an image holds the advantage on that factor.

Table 3 Mention Rate, Average Marginal Effect (AME) and Odds Ratio of leading factors

Dimension	Factor	Mention Rate (%)	AME	Odds Ratio($exp(\beta_f)$)
Accessibility	Accessibility quality	74.8	0.124	1510
	POI count	96.5	0.120	664
	Accessibility facility	9.8	0.020	105
	Land use mix	48.1	0.017	53
	Access points	12.0	0.016	57
	Connectivity	23.9	0.016	42
	Street width	19.6	0.014	40
Comfort	Psychological comfort	82.1	0.205	2898
	Building enclosure	97.6	0.114	398
	Green view rate	96.4	0.042	63
	Shade coverage	29.2	0.034	208
	Human scale	20.4	0.019	50
	Landscape diversity	27.9	0.017	36
Safety	Vegetation	16.7	0.013	25
	Visibility	95.2	0.190	1617
	Conflict risk	88.5	0.044	123
	Mixed traffic	80.4	0.019	34
	Physical separation	47.4	0.016	50
	Lane marking	59.8	0.013	35
	Pavement quality	40.6	0.005	8
	Bike lane	13.2	0.002	3

6 Discussions

Comparing a factor’s Mention Rate with its AME reveals how closely the cues the model verbalizes track the variables that drive its judgments (Figure 5). In general, Mention Rate and

AME are aligned: the factors the model mentions most frequently are also those that most significantly shift the predicted outcome. However, analyzing these two dimensions—particularly where they do not align—provides a more nuanced picture of factor importance. For instance, a factor may be universally noticed yet contribute little to the final decision (high salience, low influence), or it may be mentioned infrequently but prove highly decisive when present (low salience, high influence).

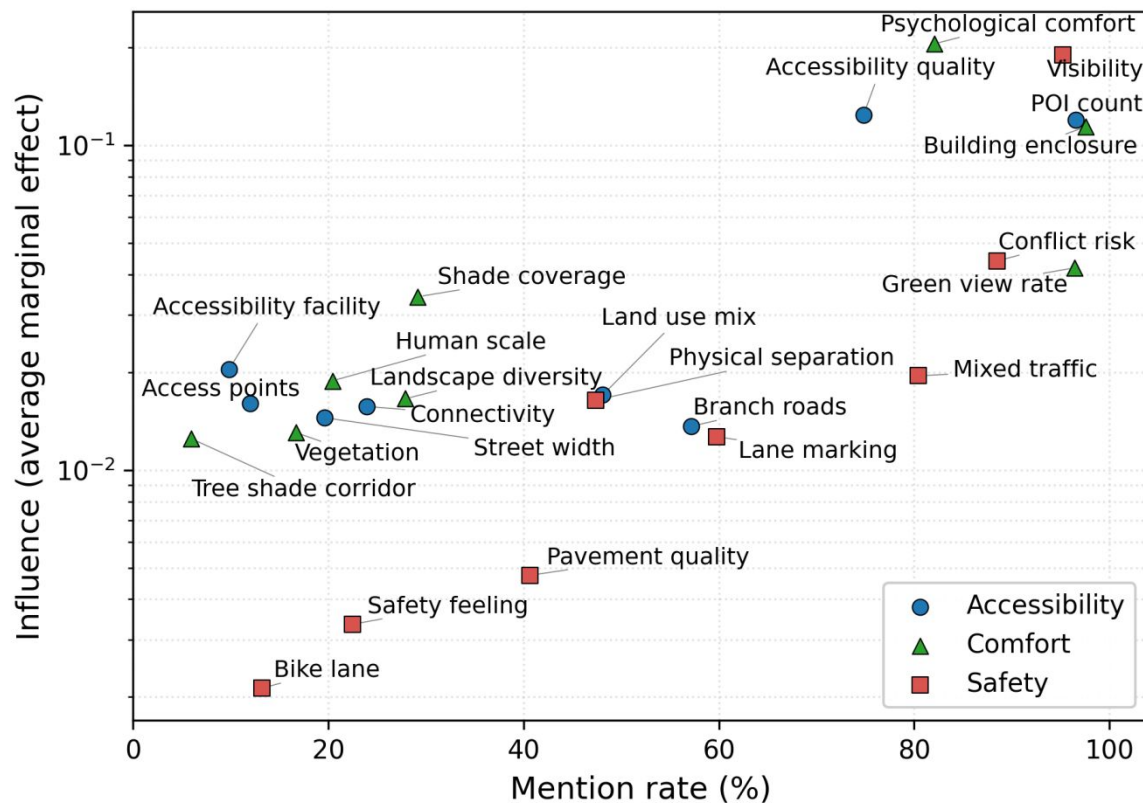


Figure 5. Mention Rate and Average Marginal Effect (AME) of street factors

A mechanistically distinct reversal appears for green view rate, which dominates Comfort judgments as a positive contributor but exhibits a significant negative effect in the Safety dimension. Dense vegetation along cycling routes can obstruct sightlines at intersections and driveways, reducing the visual openness that cyclists rely on to detect potential conflicts—consistent with Friel et al. (2023), who found that elements obstructing visibility significantly decrease cyclists' perceived safety. The implication is that greenery's contribution to bikeability is dimension-dependent: enhancing comfort through aesthetic and thermal benefits while potentially undermining safety through visual occlusion.

These Mention Rate - Influence discrepancies carry direct planning implications. First, latent but influential factors represent high-leverage intervention targets that are systematically undervalued in conventional assessment. Improving access points design, extending continuous vegetation, and enhancing connectivity may yield disproportionate returns in perceived bikeability relative to their cost and visibility—consistent with Forsyth & Krizek (2011), who argue that cycling-oriented urban design should move beyond facilities and network completeness to consider how small-scale urban design details shape the experience from the cyclist’s point of view; the influence analysis can provide a corrective lens. Readily perceptible but weakly decisive factors, by contrast, offer limited marginal perceptual returns relative to the latent factors identified above. The cross-dimensional anomalies demonstrate that cycling infrastructure planning cannot be treated as a single-objective optimization problem: interventions that enhance accessibility may inadvertently erode comfort if the spatial context is not simultaneously addressed.

Furthermore, the ability of MLLMs to capture nuanced, microscale environmental attributes marks a significant departure from traditional CV approaches. MLLMs leverage open-vocabulary vision reasoning to interpret fine-grained built environment features holistically (Ki et al., 2025). This explains why our model successfully identifies latent but highly influential factors that conventional pixel-based methods often overlook. Moreover, MLLMs offer strong zero-shot generalization across diverse urban contexts, mitigating the transferability bottlenecks that plague deep learning models trained on city-specific datasets (Zhang et al., 2025). The Mention Rate–AME framework proposed in this study provides an interpretable, dual-axis evaluation of both the model’s visual attention (what it sees) and its decision logic (what it values), thereby bridging the divide between high-perception accuracy and actionable urban insights.

7 Conclusions

This study explores the application of MLLMs to the subjective assessment of street-level bikeability and proposes a bikeability assessment framework. By supplementing street-view imagery with a specialized geographical context database by computer vision, we enable MLLMs to perform pairwise comparisons across three dimensions of bikeability. Empirically, the MLLM’s ratings demonstrate significant positive correlation with human evaluations. This study combines MLLMs with computer vision, and demonstrate its advantages. The inclusion of structured geographical context by computer vision further align model outputs with human judgment,

proving that contextual data is vital for simulating nuanced environmental perception. Future research would integrate diverse demographic variables and personality traits to capture a broader spectrum of cyclist experiences (Quintana et al., 2025).

Beyond methodological innovation, this study underscores a paradigm shift in urban perception analytics: from static, feature-extraction-based metrics to dynamic, context-aware reasoning. By demonstrating that MLLMs can reliably simulate human-like evaluative judgments across multiple perceptual dimensions, we pave the way for scalable, cost-effective urban audits that do not require extensive manual labeling or city-specific model retraining. Future research should extend this framework by integrating multi-temporal SVI data to capture seasonal and diurnal variations in urban perception. Ultimately, embedding explainable MLLM assessments into urban planning workflows will enable practitioners to move beyond broad-stroke infrastructure upgrades toward precision interventions that align with the nuanced, multi-dimensional ways people experience the city.

We have exploited the explanatory reasoning outputs to understand the logic in MLLM judgments. A key contribution of this research is the exploration of street factor salience and its influence on bikeability perception. Our analysis reveals that latent infrastructure elements—such as shade coverage and street interface connectivity—exert a disproportionately strong impact on perception despite being infrequently mentioned in reasoning texts.

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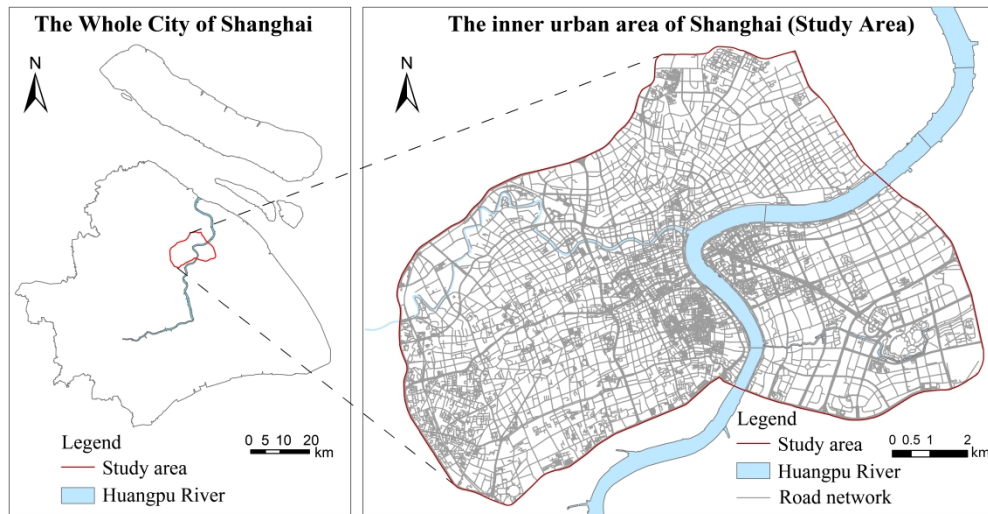


Figure 1. Inner urban area of Shanghai

819x426mm (236 x 236 DPI)

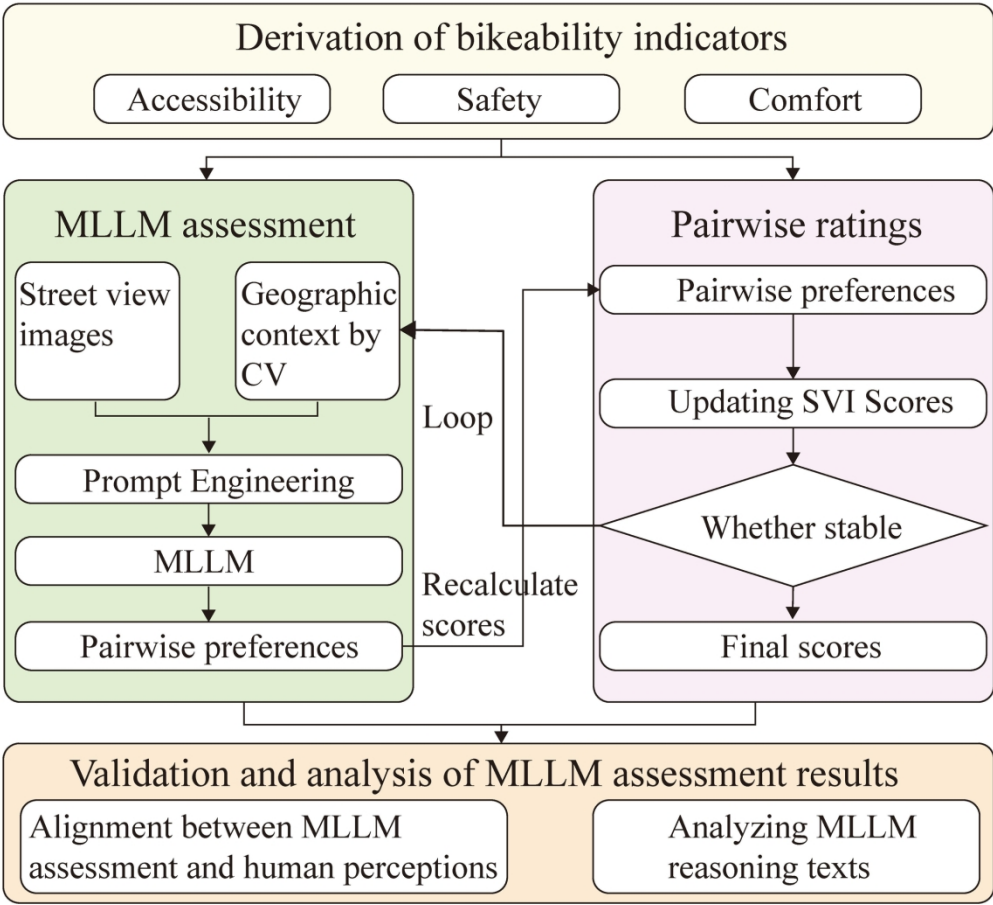


Figure 2. Analytical framework.

188x173mm (300 x 300 DPI)



Figure 3. Existing works vs. our analytical framework

861x378mm (118 x 118 DPI)

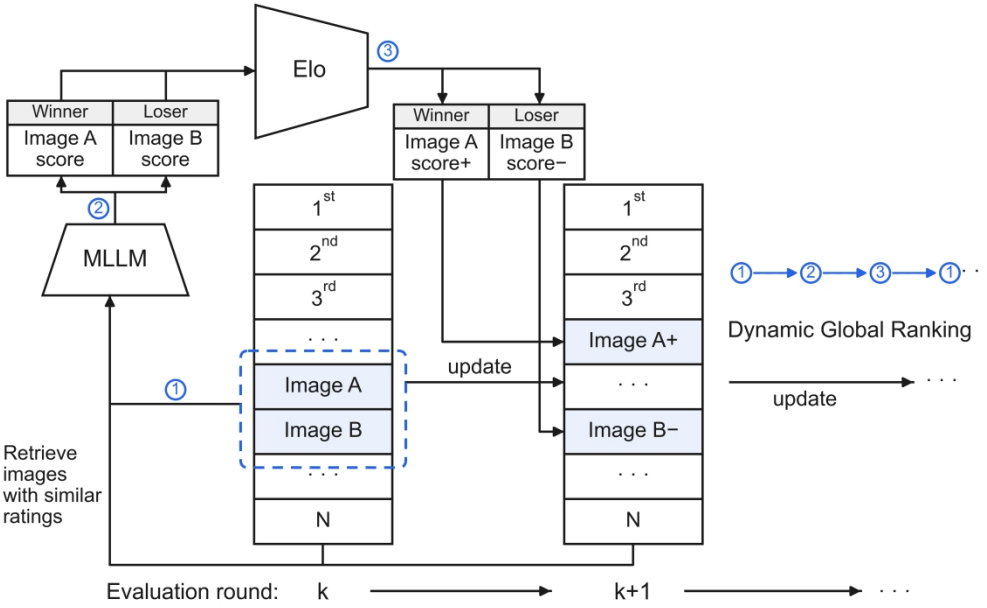


figure4.A closed adaptive loop for the MLLM pipeline

403x254mm (300 x 300 DPI)

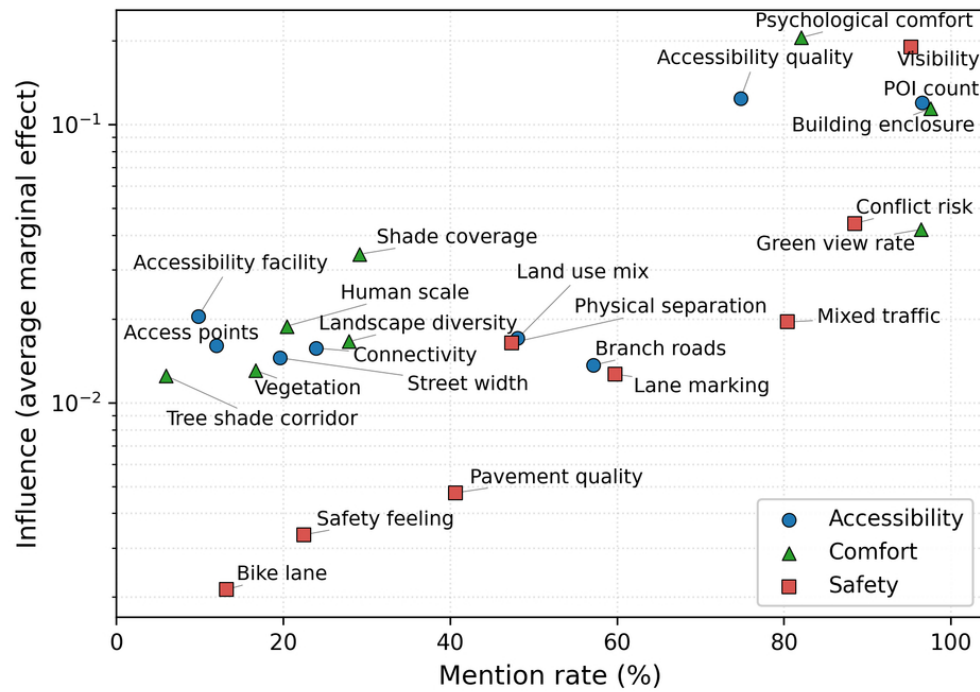


Figure 5 Mention Rate and Average Marginal Effect (AME) of street factors

79x56mm (300 x 300 DPI)

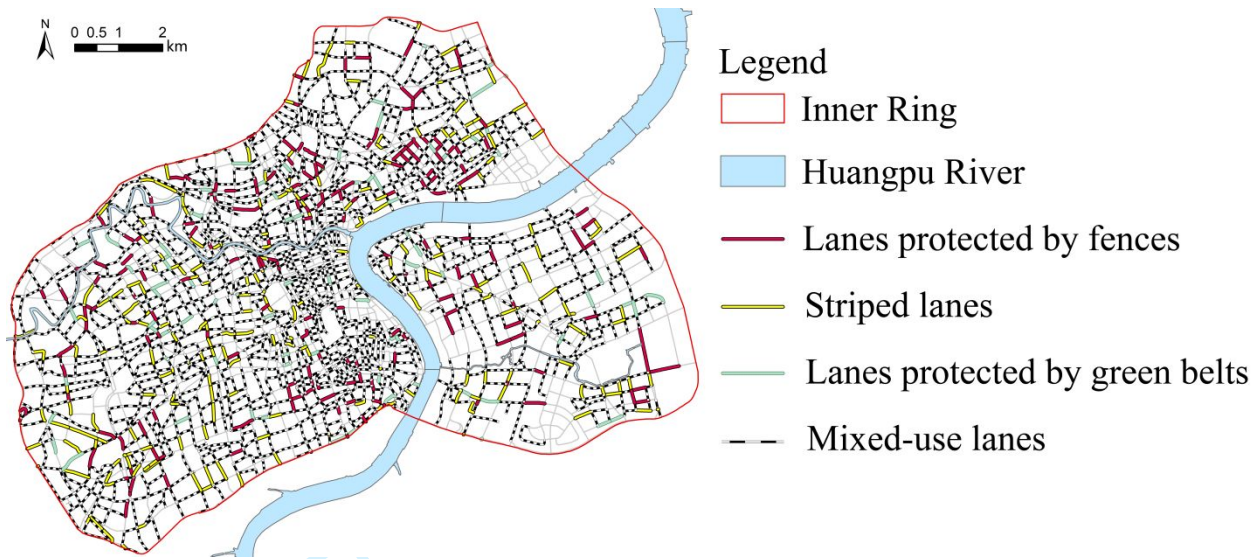


Figure S1. Object detection results of motor-nonmotor vehicle lane separation types.

Input		
Street View Images + Geographical Context Database		
	Image A	Image B
Output		
Dimension	Supporting textual Reasoning	
Accessibility Winner: image B	The geographic context data shows that Image B has 29 POIs nearby which is more than Image A with 20 ones, suggesting richer service functions . In the image, side streets on both sides connect clearly, with no obvious fences or dead ends , making it easy to enter and leave from different directions. In addition, the street interface in Image A is more enclosed where buildings are dense on both sides and there are no clear access points , which are less convenient to pass through.	
Safety Winner: image A	Image A has wide lanes with clear separation of vehicles and pedestrians, reducing conflict risks. The road surface is smooth and the view is open, so the overall traffic order is good and cycling safety is relatively high. In contrast, the road in Image B is narrow, with cars, bikes, and pedestrians mixed together. Many motorcycles and electric bikes are parked along the roadside, creating potential conflict risks . In addition, the visibility is limited, leading to greater safety concerns.	
Comfort Winner: image B	Image B offers greater environmental comfort, supported by a significantly higher Green View Index (19.2% vs. 5.9%) which provides shade and aesthetic value. Image B has a higher building enclosure ratio (2.32), which creates a street environment that feels more organized . By comparison, the lower enclosure ratio in Image A (1.41) makes the space feel open and empty , reducing the visual appeal of the street and the rider's sense of comfort.	

Figure S2. Examples of MLLM's attention to the street factors

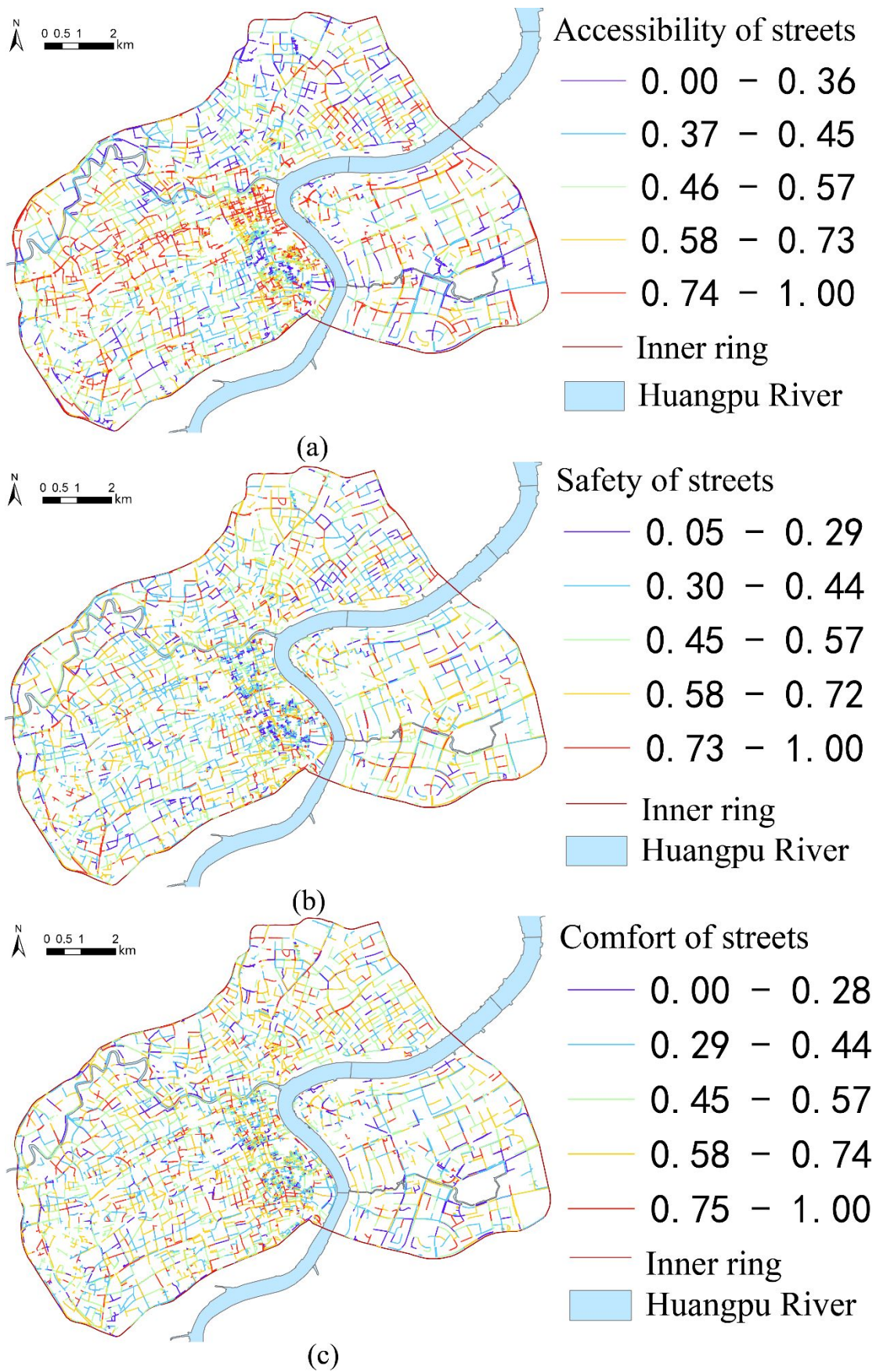


Figure S3. Spatial distribution of bikeability scores in three dimensions